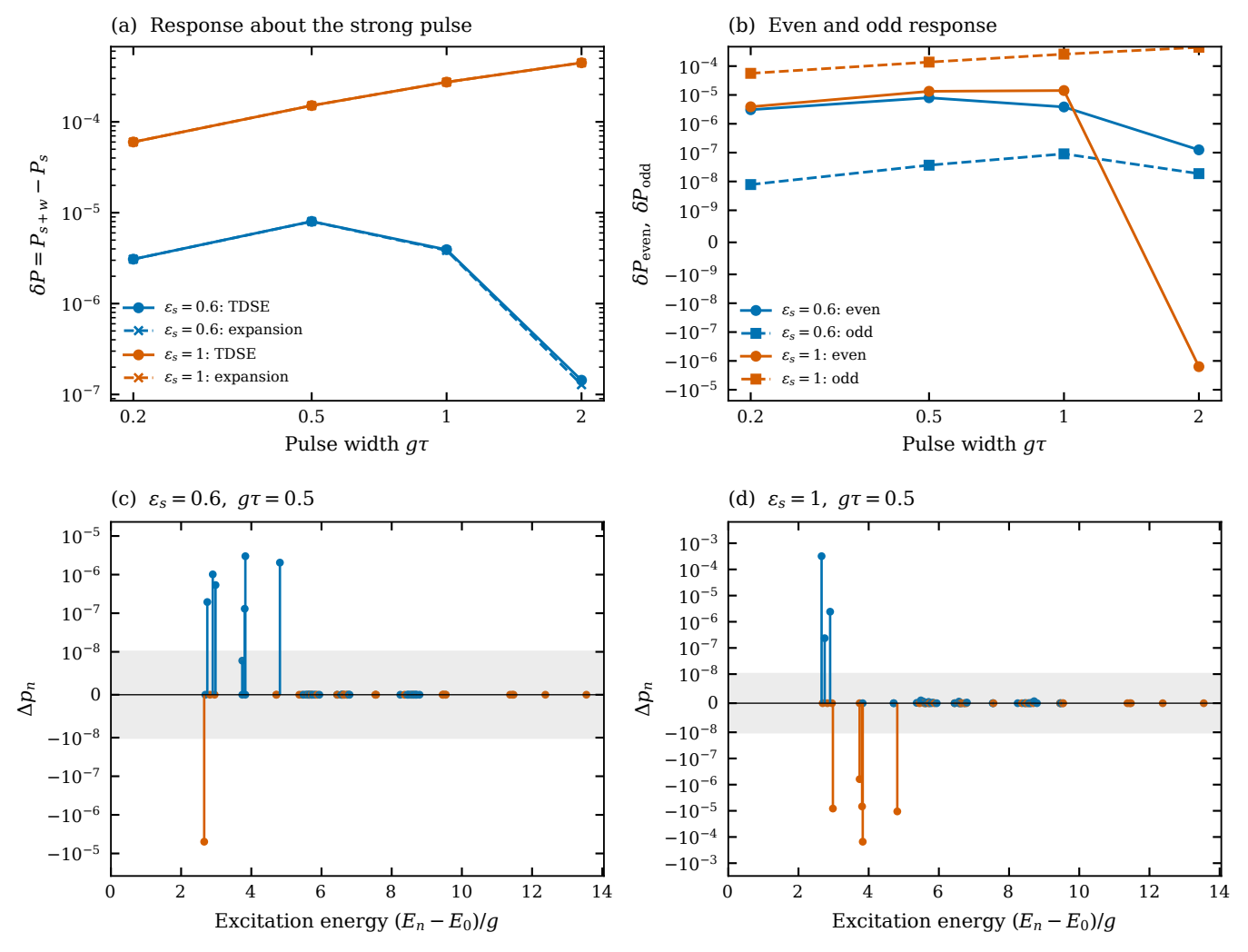
(a) Response about the strong pulse
$\delta P = P_{s+w} - P_s$
$\varepsilon_s = 0.6$: TDSE
$\varepsilon_s = 0.6$: expansion
$\varepsilon_s = 1$: TDSE
$\varepsilon_s = 1$: expansion
Pulse width $g\tau$
(b) Even and odd response
δP_{even}, δP_{odd}
$\varepsilon_s = 0.6$: even
$\varepsilon_s = 0.6$: odd
$\varepsilon_s = 1$: even
$\varepsilon_s = 1$: odd
Pulse width $g\tau$
(c) $\varepsilon_s = 0.6$, $g\tau = 0.5$
Δp_n
Excitation energy $(E_n - E_0)/g$
(d) $\varepsilon_s = 1$, $g\tau = 0.5$
Δp_n
Excitation energy $(E_n - E_0)/g$